

Five Ways to Fool Yourself About Steering a Frozen Model

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Abstract

We measure six mechanisms for changing the interpersonal style of a frozen Qwen3.6-35B-A3B, and score each one twice: on the likelihood of curated reference responses, and on text the model actually generates. The two measurements disagree, and the disagreement is the result. A residual direction trained directly against paired response likelihood drives its training objective from +0.366 to -1.188 and decorrelates its axes to a mean cross-cosine of 0.031, then moves generated text by 5.1% of the explicit-prompt span on warmth and -3.7% on patience, where the sign is reversed. Its signed ordering on held-out prompts is 4/8 and 2/8, at or below chance. A static direction and a contextual multi-site variant show smaller versions of the same split, and on one axis the two measurements disagree in sign. The teacher-forced number is the one a steering study most naturally reports, it is measured in the space the writer was optimized in, and on this evidence it certifies nothing. Only the free-generation audit with a wrong-axis control separates a writer that works from one that has learned a continuation discriminator.

1 Introduction

A steering method is usually reported by how far it moves a likelihood. The appeal is obvious: likelihood is cheap, differentiable, and available on held-out data without generating anything. The risk is equally obvious once stated. If the writer was optimized against that likelihood, then reporting it back measures how well the optimizer did its job, not whether the model’s behaviour changed.

This note makes the gap concrete. We build six writers for three signed social style axes, hold the language model frozen throughout, and score every one of them on both quantities: a teacher-forced span over curated positive and negative reference responses, and a generated span recomputed on greedily decoded text under a wrong-axis control. The mechanisms are ordered by how convincing their teacher-forced numbers are. That order does not survive contact with the second measurement.

Status. This is a confirmatory re-run. The ladder’s qualitative outcomes were established in earlier unpublished work, so its doses, layers, and step counts were selected with knowledge of those outcomes. It makes no preregistration claim, and we say so here rather than let the reader assume otherwise. What the run establishes is that the effect reproduces in a clean implementation that shares no code with the original.

2 Setup

The frozen model is Qwen3.6-35B-A3B at revision 995ad96eacd98c81ed38be0c5b274b04031597b0, loaded in bfloat16 on one NVIDIA RTX PRO 6000. The three axes are warmth versus cold hostility, patience versus annoyance, and goodwill versus resentment. The corpus is the synthetic 16-case technical-request set with content-matched styled responses, hash 85f6e3599474ad33c37092b583b21ccfc3175fd85b168c65b735c2dd75332346, split six fit, two development, eight held out. The held-out cases include repeated demands, a direct insult, praise, and gratitude.

The two measurements. For a held-out pair, the *teacher-forced span* is the positive-minus-negative reference NLL contrast under a positive intervention minus the same contrast under a negative one. The *generated span* decodes greedily under each sign and recomputes the contrast on the produced text. Both are quoted as a fraction of the span that visible style instructions achieve (Stage A), because a raw likelihood shift has no scale.

Table 1: Every writer, scored both ways. Teacher-forced spans are measured in the space the writer was optimized in; generated spans are recomputed on greedily decoded text. Percentages are of the Stage A explicit-prompt span for that axis. Signed counts held-out prompts with the intended ordering; specific counts those where the target span also beat the wrong-axis shift.

Stage	Axis	Teacher	Generated	Gen/A	Signed	Specific
D (static)	warmth	+0.0359	+0.0361	21.8%	6/8	5/8
	patience	+0.0508	+0.0636	32.2%	6/8	6/8
	goodwill	+0.0465	+0.0080	10.9%	2/8	1/8
F (contextual)	warmth	+0.0250	+0.1155	69.8%	–	–
	patience	+0.0573	+0.0953	48.2%	–	–
	goodwill	−0.0265	+0.0594	80.7%	–	–
G (trained)	warmth	*	+0.0085	5.1%	4/8	5/8
	patience	*	−0.0073	−3.7%	2/8	2/8
	goodwill	*	+0.0180	24.5%	5/8	5/8

* Stage G optimizes the teacher-forced objective directly; its training loss falls from +0.366 to −1.188, so a held-out teacher-forced span is not an independent measurement of it.

Why a wrong-axis control is not optional. The fitted axes are correlated. A large target span means nothing unless it exceeds the shift caused by steering along a different axis, so every generated measurement below reports both.

3 The ladder

Stage A: the denominator. Explicit style instructions give spans of +0.1655 (warmth), +0.1977 (patience), and +0.0737 (goodwill). Goodwill’s upper bound is weak enough that its later fractions should be read with caution; we return to this in the limitations.

Stage B: static directions. One direction per axis, component, and layer, from mean positive-minus-negative activation differences at the prompt frontier over the fit split, at residual and attention outputs of layers 19, 27, and 35.

Stage C: dose and window. A sweep over residual fractions {0.005, 0.01, 0.02, 0.04, 0.08} and injection windows of the first token, the first eight, and all response positions, scored teacher-forced. This is where a writer first looks like it works: best spans reach +0.0508 (patience), +0.0465 (goodwill), and +0.0359 (warmth).

Stage D: what the text says. Table 1 recomputes those writers on generated text. Patience and warmth survive in weakened form, at 32.2% and 21.8% of the explicit span with 6/8 signed orderings each. Goodwill does not: 10.9%, 2/8 signed, and specific on a single prompt out of eight.

Stage F: contextual, multi-site. Directions derived from the specific prompt being answered, injected coherently at all six sites under a fixed total norm budget split as $\text{total}/\sqrt{\text{sites}}$. This is a deliberately privileged upper bound rather than a deployable writer. It produces the largest generated spans in the study, and also the study’s cleanest inversion: on goodwill the teacher-forced span is −0.0265 while the generated span is +0.0594. The two measurements disagree in sign on the same writer.

Stage G: train the thing directly. One antisymmetric residual direction per axis at layer 35, initialized from Stage B, trained for 72 steps against paired full-response likelihood with a cross-axis cosine penalty. It succeeds at what it was asked to do. The loss falls from +0.366 to −1.188 and the mean absolute cross-axis cosine falls to 0.031, so the axes are no longer the single correlated style factor the fitted geometry started from. Then the generated audit returns +0.0085 on warmth (5.1%), +0.0180 on goodwill (24.5%), and −0.0073 on patience, a reversal. Signed orderings are 4/8, 5/8, and 2/8.

4 What this means for reporting steering

The ranking by teacher-forced evidence and the ranking by behaviour are not the same ranking. Stage G is the most thoroughly optimized writer here, has the cleanest axis geometry, and changes generated text least. Stage F has the weakest teacher-forced numbers of the three and the strongest generated ones, and on one axis its two scores point in opposite directions.

The mechanism is not mysterious. A direction trained against the likelihood of a fixed reference continuation can learn to discriminate that continuation from its content-matched opposite without ever making the model’s own decoding trajectory prefer it. Greedy decoding leaves the trained trajectory at the first token and never returns to it. Nothing in the teacher-forced measurement can see that happen, because the measurement never lets the model choose its own tokens.

This is the same shape as a result we reported for sensor conditioning, where a capacity-matched control reproduced a bridge’s entire bits-per-byte gain while task-aligned probes showed decisive input dependence [1]. Both are instances of one rule: a metric computed in the space an intervention was optimized in certifies nothing, and certification requires a probe outside that space. In the sensor case the probe is a task-aligned ranking; here it is free generation with a wrong-axis control.

5 Limitations

Goodwill’s explicit-prompt span, +0.0737, is less than half of patience’s, and every goodwill percentage in Table 1 is a fraction of that small denominator. An earlier internal run of the same protocol measured a substantially larger goodwill upper bound, so this axis is unstable across runs and its fractions should be treated as exploratory rather than compared against the other two.

The corpus is synthetic, English, and technical. Eight held-out prompts is a small denominator for counts like 6/8, and this run is single-seed: the companion soft-prefix study found seed-dependent behaviour severe enough that one seed refused where two others complied, so single-seed counts here should be read as indicative. Attribution is scored by the same frozen model under style instructions, which is a likelihood contrast rather than a rubric judgment, but it is not an independent judge; human evaluation would test whether the measured directions agree with reader perception. We report one dose per stage after the Stage C sweep and did not extend Stage G’s budget, so its failure is a statement about 72 steps at this dose, not about the mechanism at every budget.

6 Reproducibility

The runner is self-contained. It pins the model revision, the corpus hash, the stage configuration, seeds, and decoding, and writes every number in this note into one summary artifact. The reference run used one NVIDIA RTX PRO 6000, peaked at 66.287 GiB reserved, and completed without an out-of-memory event. Free memory gates are parameterized rather than hard preflight aborts, so the suite is portable to a different card; a 2B smoke path exercises the same code for anyone without an 80 GB-class GPU.

```
uv sync
uv run expression-ladder check
uv run expression-ladder reproduce --suite full --device cuda \
  --output-dir results/ladder-35b
uv run python -m expression_ladder.report results/ladder-35b/summary.json
```

References

- [1] F. Penov. Conditional LoRA Bridges for Modular Sensor Adaptation of Frozen Small Language Models. Preprint, 2026. <https://github.com/kortexa-ai/legolm.paper>.